



SYDNEY BOYS HIGH
MOORE PARK, SURRY HILLS

SEPTEMBER 2011
YEARLY
YEAR 11 Continuers

Mathematics Ext 1

General Instructions:

- Reading time—5 minutes.
- Working time—90 minutes.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may NOT be awarded for messy or badly arranged work.
- Answer in simplest exact form unless otherwise stated.

Total marks—68 Marks

- Attempt questions 1–4.
- The mark-value of each question is boxed in the right margin.
- Start each NEW section in a separate answer booklet.
- Hand in your answer booklets in 2 sections:
Section A (Questions 1 and 2),
Section B (Questions 3 and 4).

Examiner: Mr P. Bigelow

Section A

Marks

Question 1 (17 marks)

(a) Differentiate the following:

5

(i) $y = \frac{4}{3\sqrt{x}},$

(ii) $y = \frac{x+1}{x-2},$

(iii) $y = (2 - 3x^2)^5.$

(b) Find the equation of the normal to $y = \frac{3}{x}$ at the point where $x = -3$.

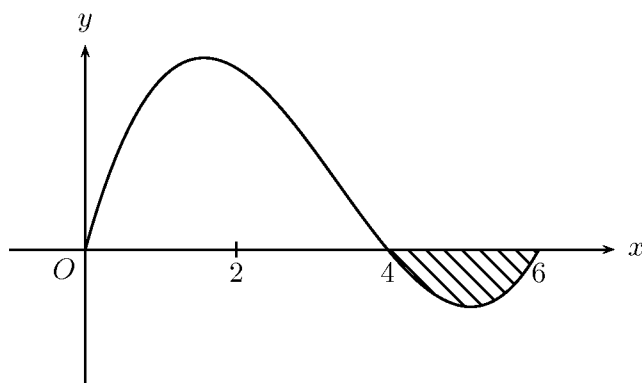
2

(c) Differentiate $y = x^2 + bx + c$, and hence find b and c if the parabola has a y -intercept of -3 and its gradient at that point is -2 .

3

(d)

2



You are given that $\int_0^4 f(x) dx = 22$ and $\int_0^6 f(x) dx = 15$.

(i) What is the area of the shaded region?

(ii) What is the value of $\int_4^6 f(x) dx$?

(e) Find $\int_1^2 (2x - 3x^2) dx$.

2

(f) θ and ϕ are acute angles and $\cos \phi = \frac{1}{3}$ and $\sin \theta = \frac{2}{3}$, find the exact value of $\cos(\theta - \phi)$.

3

Question 2 (17 marks)

(a) Consider the function $f(x) = 2 + 3x - x^3$.

7

(i) Find $f'(x)$.

(ii) Find $f''(x)$.

(iii) Locate any stationary points and determine their nature.

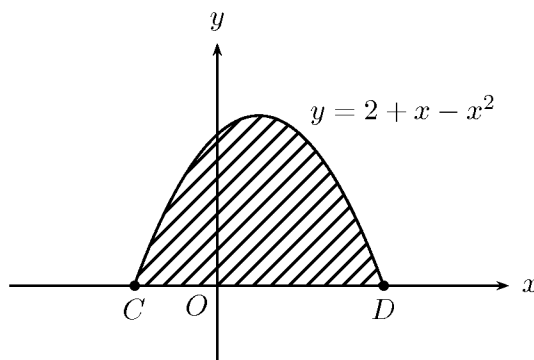
(iv) For what values of x is the curve increasing?

(v) For what values of x is the curve concave up?

(vi) Sketch the curve.

(b)

4



(i) Find the co-ordinates of C and D .

(ii) Find the ratio in which the shaded area is divided by the y -axis.

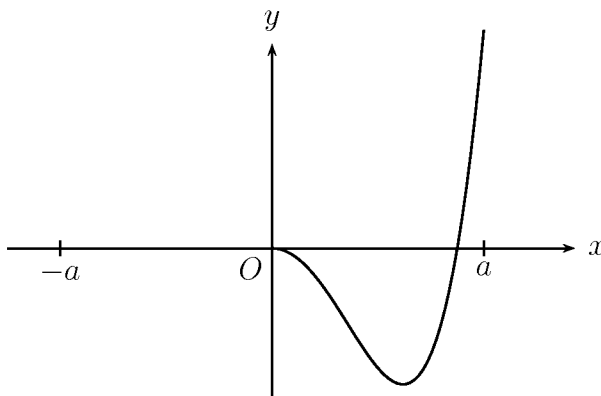
(c) (i) Expand $\sin(A + B)$.

3

(ii) Hence show that $\sin 75^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}$.

(d)

3



The diagram shows the graph of the function for $0 \leq x \leq a$.

It is known that $f(x)$ is an odd function and stationary at $(0, 0)$.

(i) Copy the diagram and continue the graph for $-a \leq x < 0$.

(ii) On another diagram, sketch $y = f'(x)$.

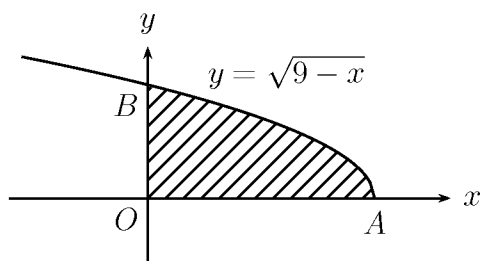
Section B

(Use a separate writing booklet.)

Marks

Question 3 (20 marks)

(a)



The shaded area is bounded by the curve $y = \sqrt{9-x}$ and the co-ordinate axes.

7

(i) Write down the values of A and B .

(ii) Find the area AOB .

(iii) Find the volume when the shaded area is rotated about

(α) the x -axis,

(β) the y -axis.

(b) Solve the following inequalities and graph the solutions on separate number lines.

4

(i) $\frac{2}{|x-1|} > 1,$

(ii) $\frac{3}{x-2} \leq \frac{2}{x+1}.$

(c) Given $\triangle ABC$ with vertices $A(3, 2)$, $B(5, 6)$ and $C(7, 4)$:

5

(i) Find the co-ordinates of E and F , the midpoints of AB and BC respectively.

(ii) Show that EF is

(α) half of

(β) and parallel to AC .

(d) Prove the identity

2

$$\frac{\sin 2A}{1 + \cos 2A} = \tan A.$$

(e) Solve $4 \sin \frac{\theta}{2} \cos \frac{\theta}{2} = 1$ for $0 \leq \theta \leq 2\pi$.

2

[HINT: use a double angle result.]

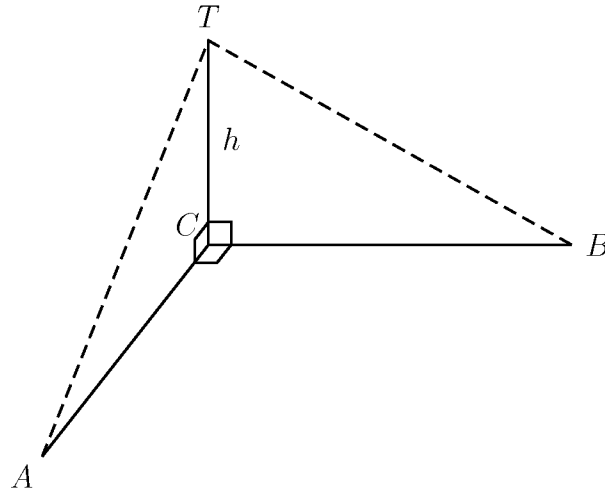
Question 4 (14 marks)

- (a) The graph of $y = f(x)$ passes through $(2, 1)$ and $f'(x) = \frac{x^3 - 4}{x^2}$. Find $f(x)$.

2

- (b)

4



CT is a tower of height h . From A , due south of the base of the tower, the angle of elevation of T , the top of the tower, is 25° . From B , due east, the angle of elevation of T is 30° . Given that AB is 200 m, find h to the nearest metre.

- (c) Let $f(x) = \frac{2x}{(1+x^2)^2}$.

8

- (i) Copy this table and complete it in decimal form.

x	1	2	3
f(x)			

- (ii) Use the table to find values for $\int_1^3 f(x) dx$ using
 (α) the Trapezoidal rule,
 (β) Simpson's rule.

- (iii) Find $\frac{d}{dx} \left(\frac{1}{x^2 + 1} \right)$.

- (iv) Using (iii), find the exact value of the integral in (ii).

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$



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Mathematics Ext 1 Solutions

Section A

Marks

Question 1 (17 marks)

(a) Differentiate the following:

5

(i) $y = \frac{4}{3\sqrt{x}},$

Solution: $y = \frac{4x^{-\frac{1}{2}}}{3}.$
 $\frac{dy}{dx} = -\frac{1}{2} \cdot \frac{4x^{-\frac{3}{2}}}{3},$
 $= -\frac{2}{3x\sqrt{x}}.$

(ii) $y = \frac{x+1}{x-2},$

Solution: $\frac{dy}{dx} = \frac{(x-2) \times 1 - (x+1) \times 1}{(x-2)^2},$
 $= \frac{-3}{(x-2)^2}.$

(iii) $y = (2 - 3x^2)^5.$

Solution: $\frac{dy}{dx} = 5 \times (-3 \times 2x) \times (2 - 3x^2)^4,$
 $= -30x(2 - 3x^2)^4.$

(b) Find the equation of the normal to $y = \frac{3}{x}$ at the point where $x = -3$.

2

Solution: $\frac{dy}{dx} = -\frac{3}{x^2},$
 $= -\frac{1}{3}$ at the point $(-3, -1).$

Normal equation:

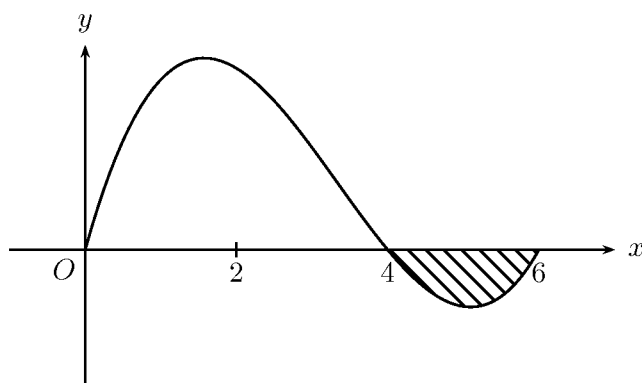
$y + 1 = 3(x + 3),$
i.e., $3x - y + 8 = 0.$

(c) Differentiate $y = x^2 + bx + c$, and hence find b and c if the parabola has a y -intercept of -3 and its gradient at that point is -2 .

3

Solution: $y' = 2x + b.$
 Now, at $(0, -3)$:
 $-2 = 2 \times 0 + b,$
i.e., $b = -2,$
 and $-3 = 0^2 - 2 \times 0 + c,$
i.e., $c = -3.$

(d)



2

You are given that $\int_0^4 f(x) dx = 22$ and $\int_0^6 f(x) dx = 15$.

(i) What is the area of the shaded region?

Solution: Shaded area = $|15 - 22|$,
 $= 7$.

(ii) What is the value of $\int_4^6 f(x) dx$?

Solution: $\int_4^6 f(x) dx = \int_0^6 f(x) dx - \int_0^4 f(x) dx$,
 $= -7$.

(e) Find $\int_1^2 (2x - 3x^2) dx$.

2

Solution: $\int_1^2 (2x - 3x^2) dx = \left[x^2 - x^3 \right]_1^2$,
 $= 4 - 8 - (1 - 1)$,
 $= -4$.

(f) θ and ϕ are acute angles and $\cos \phi = \frac{1}{3}$ and $\sin \theta = \frac{2}{3}$, find the exact value of $\cos(\theta - \phi)$.

3

Solution:

$$\begin{aligned} \cos(\theta - \phi) &= \cos \theta \cos \phi + \sin \theta \sin \phi, \\ &= \frac{\sqrt{5}}{3} \cdot \frac{1}{3} + \frac{2}{3} \cdot \frac{2\sqrt{2}}{3}, \\ &= \frac{\sqrt{5} + 4\sqrt{2}}{9}. \end{aligned}$$

Question 2 (17 marks)(a) Consider the function $f(x) = 2 + 3x - x^3$.

7

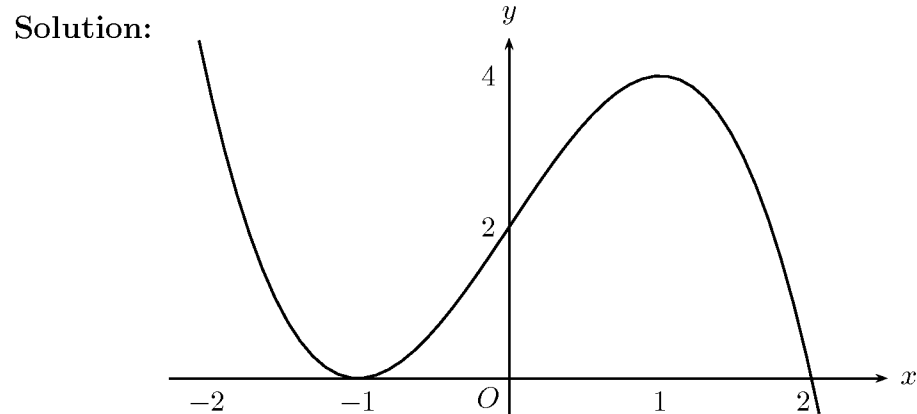
(i) Find $f'(x)$.**Solution:** $f'(x) = 3 - 3x^2$.(ii) Find $f''(x)$.**Solution:** $f''(x) = -6x$.

(iii) Locate any stationary points and determine their nature.

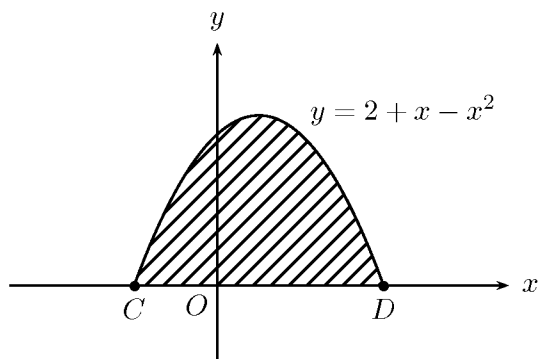
Solution: When $f'(x) = 0$, and when $f''(x) = 0$,
 $3(1 - x^2) = 0$, $x = 0$, $y = 2$,
 $x = \pm 1$, $f''(1) = -6$,
 $y = 4, 0$. $f''(-1) = 6$.
 $\therefore$ A maximum at $(1, 4)$ and a minimum at $(-1, 0)$.

(iv) For what values of x is the curve increasing?**Solution:** $-1 < x < 1$.(v) For what values of x is the curve concave up?**Solution:** $x < 0$.

(vi) Sketch the curve.



(b)

(i) Find the co-ordinates of C and D .

Solution: $x^2 - x - 2 = 0,$
 $(x - 2)(x + 1) = 0,$
 $\therefore x = -1, 2.$
i.e., $C(-1, 0)$ and $D(2, 0).$

(ii) Find the ratio in which the shaded area is divided by the y -axis.

Solution: $\int_{-1}^0 (2 + x - x^2) dx = \left[2x + \frac{x^2}{2} - \frac{x^3}{3} \right]_{-1}^0,$
 $= 0 - \left\{ -2 + \frac{1}{2} - \frac{1}{3} \right\},$
 $= 1\frac{5}{6}.$
 $\int_0^2 (2 + x - x^2) dx = \left[2x + \frac{x^2}{2} - \frac{x^3}{3} \right]_0^2,$
 $= \left\{ 4 + \frac{4}{2} - \frac{8}{3} \right\} - 0,$
 $= 5.$
i.e., the ratio is 11 : 30.

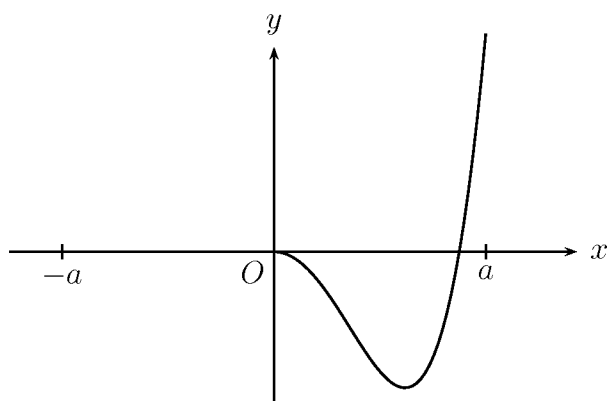
(c) (i) Expand $\sin(A + B)$.

Solution: $\sin(A + B) = \sin A \cos B + \cos A \sin B.$

(ii) Hence show that $\sin 75^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}.$

Solution: $\sin(45^\circ + 30^\circ) = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ,$
 $= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \times \frac{1}{2},$
 $= \frac{\sqrt{6} + \sqrt{2}}{4}.$

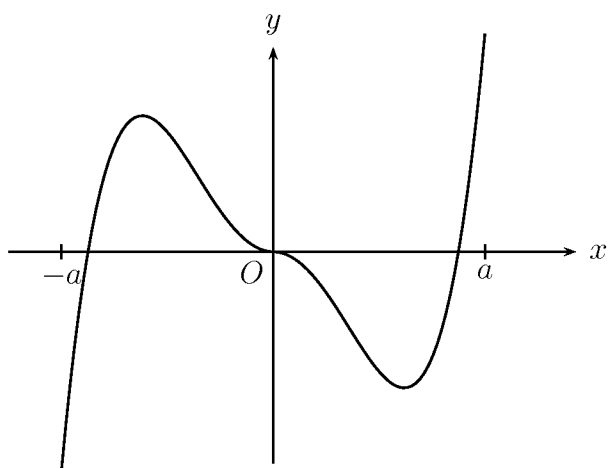
(d)



The diagram shows the graph of the function for $0 \leq x \leq a$.
It is known that $f(x)$ is an odd function and stationary at $(0, 0)$.

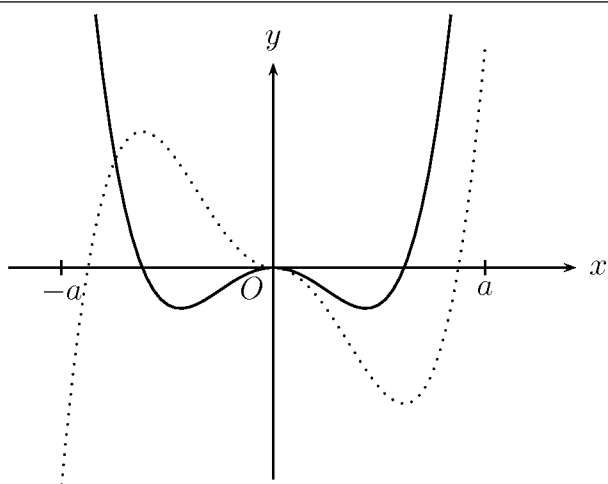
(i) Copy the diagram and continue the graph for $-a \leq x < 0$.

Solution:



(ii) On another diagram, sketch $y = f'(x)$.

Solution:



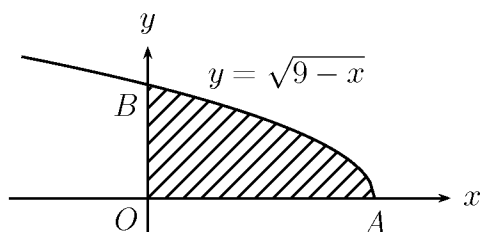
Section B

(Use a separate writing booklet.)

Marks

Question 3 (20 marks)

(a)



The shaded area is bounded by the curve $y = \sqrt{9-x}$ and the co-ordinate axes.

7

(i) Write down the values of A and B .

Solution: $A(9, 0)$, $B(0, 3)$

(ii) Find the area AOB .

Solution:
$$\int_0^9 (9-x)^{\frac{1}{2}} dx = \left[-1 \times \frac{2}{3} \times (9-x)^{\frac{3}{2}} \right]_0^9,$$
$$= -\frac{2}{3}\{0 - 27\},$$
$$= 18.$$

(iii) Find the volume when the shaded area is rotated about
(α) the x -axis,

Solution: Note that $y^2 = 9 - x$.

$$\begin{aligned} \text{Volume} &= \pi \int_0^9 (9-x) dx, \\ &= \pi \left[9x - \frac{x^2}{2} \right]_0^9, \\ &= \pi \left\{ 81 - \frac{81}{2} - 0 \right\}, \\ &= \frac{81\pi}{2}. \end{aligned}$$

(β) the y -axis.

Solution: Note: $y^2 = 9 - x$,

$$x = 9 - y^2,$$

$$x^2 = 81 - 18y^2 + y^4.$$

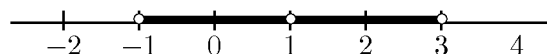
$$\begin{aligned} \text{So Volume} &= \pi \int_0^3 (81 - 18y^2 + y^4) dy, \\ &= \pi \left[81y - 6y^3 + \frac{y^5}{5} \right]_0^3, \\ &= \pi \left\{ 243 - 162 + \frac{243}{5} \right\}, \\ &= \frac{648\pi}{5}. \end{aligned}$$

(b) Solve the following inequalities and graph the solutions on separate number lines.

4

(i) $\frac{2}{|x-1|} > 1,$

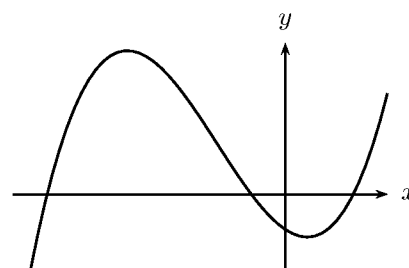
Solution: $2 > |x-1|,$
 $-2 < x-1 < 2,$
 $-1 < x < 3, x \neq 1.$



(ii) $\frac{3}{x-2} \leq \frac{2}{x+1}.$

Solution: $\frac{3}{x-2} - \frac{2}{x+1} \leq 0,$
 $\frac{3(x+1) - 2(x-2)}{(x-2)(x+1)} \leq 0,$
 $\frac{x+7}{(x-2)(x+1)} \leq 0,$
 $(x+7)(x-2)(x+1) \leq 0.$

We note that $x \neq -1$ and $x \neq 2$,
 $\therefore x \leq -7, -1 < x < 2.$



(c) Given $\triangle ABC$ with vertices $A(3, 2)$, $B(5, 6)$ and $C(7, 4)$:

5

(i) Find the co-ordinates of E and F , the midpoints of AB and BC respectively.

Solution: E is $\left(\frac{3+5}{2}, \frac{2+6}{2}\right) = (4, 4),$
 F is $\left(\frac{5+7}{2}, \frac{6+4}{2}\right) = (6, 5).$

(ii) Show that EF is
 (α) half of

Solution: $AC^2 = (3-5)^2 + (2-6)^2,$
 $= 20,$
 $AC = 2\sqrt{5}.$
 $EF^2 = (4-6)^2 + (4-5)^2,$
 $= 5,$
 $EF = \sqrt{5} = \frac{1}{2}AC.$

(β) and parallel to AC .

Solution: Slope, $m_{AC} = \frac{2-4}{3-5} = \frac{1}{2}.$
 $m_{EF} = \frac{4-5}{4-6} = \frac{1}{2}. \quad \text{i.e., } EF \parallel AC.$

(d) Prove the identity

2

$$\frac{\sin 2A}{1 + \cos 2A} = \tan A.$$

Solution:

$$\begin{aligned}\text{L.H.S.} &= \frac{2 \sin A \cos A}{1 + 2 \cos^2 A - 1}, \\ &= \frac{\cancel{2} \sin A \cancel{\cos A}}{\cancel{2} \cos^2 A}, \\ &= \frac{\sin A}{\cos A}, \\ &= \tan A, \\ &= \text{R.H.S.}\end{aligned}$$

(e) Solve $4 \sin \frac{\theta}{2} \cos \frac{\theta}{2} = 1$ for $0 \leq \theta \leq 2\pi$.
[HINT: use a double angle result.]

2

Solution:

$$\begin{aligned}4 \sin \frac{\theta}{2} \cos \frac{\theta}{2} &= 2 \sin \theta. \\ \text{So } \sin \theta &= \frac{1}{2}, \\ \theta &= \frac{\pi}{6}, \frac{5\pi}{6}.\end{aligned}$$

Question 4 (14 marks)

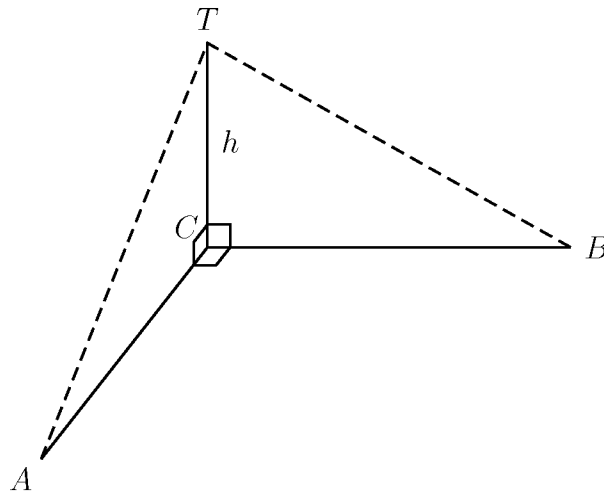
- (a) The graph of $y = f(x)$ passes through $(2, 1)$ and $f'(x) = \frac{x^3 - 4}{x^2}$. Find $f(x)$. 2

Solution:

$$\begin{aligned} \int \frac{x^3 - 4}{x^2} dx &= \int (x - 4x^{-2}) dx, \\ &= \frac{x^2}{2} - \frac{4x^{-1}}{-1} + c, \\ &= \frac{x^2}{2} + \frac{4}{x} + c. \end{aligned}$$

So at $(2, 1)$ we have $1 = \frac{2^2}{2} + \frac{4}{2} + c$,
i.e., $c = -3$.
Hence $f(x) = \frac{x^2}{2} + \frac{4}{x} - 3$.

- (b) 4



CT is a tower of height h . From A , due south of the base of the tower, the angle of elevation of T , the top of the tower, is 25° . From B , due east, the angle of elevation of T is 30° . Given that AB is 200 m, find h to the nearest metre.

Solution:

$$\begin{aligned} \frac{h}{AC} &= \tan 25^\circ \implies AC = h \cot 25^\circ, \\ \frac{h}{BC} &= \tan 30^\circ \implies BC = h \cot 30^\circ. \end{aligned}$$

$$\begin{aligned} AC^2 + BC^2 &= 200^2, \\ h^2 \cot^2 25^\circ + h^2 \cot^2 30^\circ &= 40\,000, \\ h^2 &= \frac{40\,000}{\cot^2 25^\circ + \cot^2 30^\circ}, \\ &\approx 5263.912898. \\ \therefore h &\approx 73 \text{ m (nearest metre).} \end{aligned}$$

(c) Let $f(x) = \frac{2x}{(1+x^2)^2}$.

(i) Copy this table and complete it in decimal form.

x	1	2	3
$f(x)$			

Solution:

x	1	2	3
$f(x)$	0.5	0.16	0.06

(ii) Use the table to find values for $\int_1^3 f(x) dx$ using

(α) the Trapezoidal rule,

Solution:
$$\int_1^3 f(x) dx \approx \frac{1}{2} \{0.5 + 2 \times 0.16 + 0.06\},$$

$$\approx 0.44.$$

(β) Simpson's rule.

Solution:
$$\int_1^3 f(x) dx \approx \frac{1}{3} \{0.5 + 4 \times 0.16 + 0.06\},$$

$$\approx 0.4.$$

(iii) Find $\frac{d}{dx} \left(\frac{1}{x^2 + 1} \right)$.

Solution:
$$\frac{d}{dx} ((x^2 + 1)^{-1}) = -1 \times 2x(x^2 + 1)^{-2},$$

$$= \frac{-2x}{(1 + x^2)^2}.$$

(iv) Using (iii), find the exact value of the integral in (ii).

Solution:
$$\int_1^3 f(x) dx = - \int_1^3 \frac{-2x}{(1 + x^2)^2} dx,$$

$$= - \left[\frac{1}{x^2 + 1} \right]_1^3,$$

$$= - \left\{ \frac{1}{10} - \frac{1}{2} \right\},$$

$$= 0.4.$$

End of Paper

